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Seal member for mounting within a waterproof connector

Abstract

A waterproof connector comprises a terminal, a housing and a seal member. The housing is provided with an attachment portion. When the attachment portion is viewed along a front-rear direction, the attachment portion has an outer circumferential shape including first sides and first coupling portions. The seal member is formed of elastic material and is attached to the attachment portion. When the seal member is viewed along the front-rear direction, the seal member has a closed-ring shape including second sides and second coupling portions. The second sides correspond to the first sides, respectively. The second sides include at least one specific second side. $L_{sub.1} > L_{sub.2}$, where $L_{sub.2}$ is a length of each of the second sides in a state before the seal member is attached to the attachment portion, and $L_{sub.1}$ is a length of the corresponding first side.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority under 35 U.S.C. § 119 to Japanese Patent

Application No. JP2021-173715 filed Oct. 25, 2021, the contents of which are incorporated herein in their entirety by reference.

BACKGROUND OF THE INVENTION

(2) This invention relates to a waterproof connector comprising a seal member.

(3) JPA2014-99255 (Patent Document 1) discloses a waterproof connector 900 of this type.

Referring to FIG. 17, the waterproof connector 900 of Patent Document 1 is mateable with a mating connector (not shown) along a mating direction, or along an X-direction. The waterproof connector 900 comprises terminals 910, a housing 920 and a seal member 950. The terminals 910 are held by the housing 920. The housing 920 is provided with an attachment portion 922. The seal member 950 is formed of elastic material and is attached to the attachment portion 922.

(4) The waterproof connector 900 of Patent Document 1 has a drawback that the seal member 950 might be twisted when the waterproof connector 900 is mated with or removed from the mating connector.

SUMMARY OF THE INVENTION

(5) It is an object of the present invention to provide a waterproof connector which can prevent twisting of a seal member when the waterproof connector is mated with or removed from a mating connector.

(6) In an attempt by the inventors of the present application to design a waterproof connector that can prevent twisting of a seal member, the inventors have found that twisting of a seal member can be prevented in a specific case where a shape and/or size of the seal member is smaller than a shape and/or size of an attachment portion of a housing in a plane perpendicular to a mating direction. In addition, the inventors also have found that sealing characteristics between a waterproof connector and a mating connector might be degraded in the specific case.

(7) Then, the inventors further studied a cause of the degradation of the sealing characteristics and have found that the degradation of the sealing characteristics is caused as follows: if amounts of distortions produced in sides of a seal member are different from one another upon attachment of the seal member to an attachment portion, forces applied to a coupling portion coupling two of the sides with each other are unbalanced, and thereby a position of the coupling portion on the attachment portion is deviated from an appropriate position.

(8) Based on the above, the inventors have conceived that a position of a coupling portion of a seal member on an attachment portion is prevented from being deviated from an appropriate position by providing recesses on opposite ends in a mating direction of its specific side, whose amount of distortion is expected to be maximum as compared to any of amounts of distortions of its sides other than the specific side, so that the specific side is easily stretched while forces applied to the coupling portion are well balanced upon attachment of the seal member to the attachment portion. The present invention stems from the conceiving as described above.

(9) One aspect of the present invention provides a waterproof connector mateable with a mating connector along a front-rear direction. The waterproof connector comprises a terminal, a housing and a seal member. The terminal is held by the housing. The housing is provided with an attachment portion. When the attachment portion is viewed along the front-rear direction, the attachment portion has an outer circumferential shape including first sides and first coupling portions. The first coupling portions extend in directions which are different from one another. Each of the first coupling portions couples the first sides, which are adjacent to each other, with each other. The seal member is formed of elastic material and is attached to the attachment portion. When the seal member is viewed along the front-rear direction, the seal member has a closed-ring shape including second sides and second coupling portions. The second sides correspond to the first sides, respectively. The second coupling portions extend in directions which are different from one another. Each of the second coupling portions couples the second sides, which are adjacent to each other, with each other. The second sides include at least one specific second side. The at least one specific second side is provided with a front recess and a rear recess. The front recess is

recessed rearward in the front-rear direction. The rear recess is recessed forward in the front-rear direction. $L_{sub.1} > L_{sub.2}$, where $L_{sub.2}$ is a length of each of the second sides in a state before the seal member is attached to the attachment portion, and $L_{sub.1}$ is a length of the corresponding first side. An expected amount G of distortion of each of the second sides upon attachment of the seal member to the attachment portion is defined by an equation of $G = (L_{sub.1} - L_{sub.2}) / L_{sub.2}$. The expected amount G of the distortion of the at least one specific second side is maximum among the expected amounts G of the distortions of the second sides.

(10) The waterproof connector of the present invention is configured as follows: when the attachment portion is viewed along the front-rear direction, the attachment portion has the outer circumferential shape including the first sides and the first coupling portions; when the seal member is viewed along the front-rear direction, the seal member has the closed-ring shape including the second sides and the second coupling portions; and $L_{sub.1} > L_{sub.2}$, where $L_{sub.2}$ is the length of each of the second sides in the state before the seal member is attached to the attachment portion, and $L_{sub.1}$ is the length of the corresponding first side. Thus, the waterproof connector of the present invention can prevent twisting of the seal member when the waterproof connector is mated with or removed from the mating connector.

(11) Furthermore, the waterproof connector of the present invention is configured as follows: the second sides include the at least one specific second side; the expected amount G of the distortion of each of the second sides upon attachment of the seal member to the attachment portion is defined by the equation of $G = (L_{sub.1} - L_{sub.2}) / L_{sub.2}$, and the expected amount G of the distortion of the at least one specific second side is maximum among the expected amounts G of the distortions of the second sides; and the at least one specific second side is provided with the front recess and the rear recess. Thus, the at least one specific second side is easily stretched as compared to any of the second sides other than the specific second side, and thereby a position of the second coupling portion on the attachment portion is prevented from being deviated from an appropriate position. That is, the waterproof connector of the present invention prevents degradation of sealing characteristics.

(12) An appreciation of the objectives of the present invention and a more complete understanding of its structure may be had by studying the following description of the preferred embodiment and by referring to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view showing a waterproof connector according to an embodiment of the present invention.

(2) FIG. 2 is a front view showing the waterproof connector of FIG. 1.

(3) FIG. 3 is a cross-sectional view showing the waterproof connector of FIG. 2, taken along line A-A.

(4) FIG. 4 is a cross-sectional view showing the waterproof connector of FIG. 2, taken along line B-B.

(5) FIG. 5 is a top view showing the waterproof connector of FIG. 1.

(6) FIG. 6 is a cross-sectional view showing the waterproof connector of FIG. 5, taken along line C-C.

(7) FIG. 7 is an exploded, perspective view showing the waterproof connector of FIG. 1.

(8) FIG. 8 is a perspective view showing a housing which is included in the waterproof connector of FIG. 7.

(9) FIG. 9 is a front view showing the housing of FIG. 8.

(10) FIG. 10 is a top view showing the housing of FIG. 8.

- (11) FIG. 11 is a cross-sectional view showing the housing of FIG. 10, taken along line D-D.
- (12) FIG. 12 is a side view showing the housing of FIG. 8.
- (13) FIG. 13 is a front view showing a seal member which is included in the waterproof connector of FIG. 7.
- (14) FIG. 14 is a top view showing the seal member of FIG. 13.
- (15) FIG. 15 is a side view showing the seal member of FIG. 13.
- (16) FIG. 16 is a rear view showing a front retainer which is included in the waterproof connector of FIG. 7.
- (17) FIG. 17 is a perspective view showing a waterproof connector of Patent Document 1.
- (18) While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the present invention as defined by the appended claims.

DETAILED DESCRIPTION

- (19) Referring to FIG. 1, a waterproof connector **100** according to an embodiment of the present invention is mateable with a mating connector along a front-rear direction. More specifically, the waterproof connector **100** of the present embodiment is mateable from its rear along the front-rear direction with the mating connector which is positioned forward of the waterproof connector **100**. In the present embodiment, the front-rear direction is an X-direction. Specifically, forward is a positive X-direction while rearward is a negative X-direction.
- (20) As shown in FIG. 7, the waterproof connector **100** of the present embodiment comprises a housing **300**, a seal member **500** and a plurality of terminals **200**. However, the present invention is not limited thereto, but the number of the terminal **200** may be one. In other words, the waterproof connector **100** should comprise the housing **300**, the seal member **500** and the terminal **200**.
- (21) Referring to FIG. 7, the housing **300** of the present embodiment is made of insulator. The housing **300** has a surrounding portion **305**.
- (22) Referring to FIG. 9, the surrounding portion **305** of the present embodiment defines an outer end of the housing **300** in a direction perpendicular to the front-rear direction.
- (23) As shown in FIG. 11, the housing **300** is provided with an attachment portion **310**.
- (24) Referring to FIG. 11, a part of the attachment portion **310** of the present embodiment is surrounded by the surrounding portion **305** in a plane perpendicular to the front-rear direction. The attachment portion **310** has an outer circumference having a rectangular shape with rounded corners in the plane perpendicular to the front-rear direction. Specifically, when the attachment portion **310** is viewed along the front-rear direction, the attachment portion **310** has an outer circumferential shape including first sides **320** and first coupling portions **330**.
- (25) As shown in FIG. 11, the first sides **320** of the present embodiment include two long sides **322** and two short sides **324**. Specifically, the first sides **320** are composed of the two long sides **322** and the two short sides **324**.
- (26) As shown in FIG. 11, each of the long sides **322** of the present embodiment extends in a horizontal direction perpendicular to the front-rear direction. In the present embodiment, the horizontal direction is a Y-direction. In addition, the horizontal direction is also referred to as a right-left direction. Specifically, it is assumed that rightward is a positive Y-direction while leftward is a negative Y-direction. The two long sides **322** are positioned away from each other in an up-down direction perpendicular to both the front-rear direction and the horizontal direction. The long side **322**, which is positioned at an upper side of the attachment portion **310**, of the two long sides **322** defines an upper end of the attachment portion **310** in the up-down direction. The long side **322**, which is positioned at a lower side of the attachment portion **310**, of the two long sides **322** defines a lower end of the attachment portion **310** in the up-down direction. In the present

embodiment, the up-down direction is a Z-direction. Specifically, upward is a positive Z-direction while downward is a negative Z-direction.

(27) As shown in FIG. 11, each of the short sides **324** of the present embodiment extends in the up-down direction perpendicular to both the front-rear direction and the horizontal direction. The two short sides **324** are positioned away from each other in the horizontal direction. The short side **324**, which is positioned at a right side of the attachment portion **310**, of the two short sides **324** defines a right end of the attachment portion **310** in the right-left direction. The short side **324**, which is positioned at a left side of the attachment portion **310**, of the two short sides **324** defines a left end of the attachment portion **310** in the right-left direction.

(28) As shown in FIG. 11, a length L.sub.1A of the long side **322** in the horizontal direction is greater than a length L.sub.1B of the short side **324** in the up-down direction.

(29) As shown in FIG. 11, the first coupling portions **330** of the present embodiment extend in directions which are different from one another. Each of the first coupling portions **330** couples the first sides **320**, which are adjacent to each other, with each other. More specifically, each of the first coupling portions **330** couples the long side **322** and the short side **324**, which are adjacent to each other, with each other.

(30) As shown in FIG. 11, the housing **300** has a rear end portion **340**.

(31) As shown in FIG. 11, the rear end portion **340** of the present embodiment faces forward in the front-rear direction. The rear end portion **340** is positioned outside the attachment portion **310** in the plane perpendicular to the front-rear direction. As understood from FIGS. 3 and 4, the attachment portion **310** extends forward in the front-rear direction from the rear end portion **340**. As shown in FIG. 3, a rear end **510** of the seal member **500** in the front-rear direction is pressed against the rear end portion **340**.

(32) As shown in FIG. 11, the housing **300** further has a wall portion **360** and a first accommodating portion **370**.

(33) As shown in FIG. 11, the wall portion **360** of the present embodiment is positioned inside the surrounding portion **305** in the direction perpendicular to the front-rear direction. The attachment portion **310** is positioned inside the wall portion **360** in the direction perpendicular to the front-rear direction. As shown in FIG. 8, the wall portion **360** extends forward in the front-rear direction from the rear end portion **340**. In the front-rear direction, a front end of the wall portion **360** is positioned rearward beyond a front end of the surrounding portion **305**. In the front-rear direction, the front end of the wall portion **360** is positioned rearward beyond a front end of the attachment portion **310**.

(34) As shown in FIG. 11, the wall portion **360** includes a right wall portion **362** and a left wall portion **364**.

(35) As shown in FIG. 11, the right wall portion **362** of the present embodiment is positioned rightward of the left wall portion **364** in the right-left direction. The right wall portion **362** and the left wall portion **364** are arranged in the right-left direction. The right wall portion **362** and the left wall portion **364** are positioned at positions same as each other in the front-rear direction.

(36) As shown in FIG. 11, the left wall portion **364** of the present embodiment is positioned leftward of the right wall portion **362** in the right-left direction.

(37) Referring to FIG. 11, the first accommodating portion **370** of the present embodiment is a recess which is recessed rearward in the front-rear direction. The first accommodating portion **370** is positioned between the wall portion **360** and the attachment portion **310**. The first accommodating portion **370** is positioned inside the wall portion **360** in the direction perpendicular to the front-rear direction. The first accommodating portion **370** is positioned outside the attachment portion **310** in the direction perpendicular to the front-rear direction. The rear end portion **340** defines a rear end of the first accommodating portion **370** in the front-rear direction. The first accommodating portion **370** includes a first right accommodating portion **372** and a first left accommodating portion **374**.

(38) As shown in FIG. 11, the first right accommodating portion **372** of the present embodiment is positioned rightward of the first left accommodating portion **374** in the right-left direction. The first right accommodating portion **372** is positioned leftward of the right wall portion **362** in the right-left direction. The first right accommodating portion **372** and the first left accommodating portion **374** are arranged in the right-left direction.

(39) As shown in FIG. 11, the first left accommodating portion **374** of the present embodiment is positioned leftward of the first right accommodating portion **372** in the right-left direction. The first left accommodating portion **374** is positioned rightward of the left wall portion **364** in the right-left direction.

(40) As shown in FIG. 11, the housing **300** is provided with two rear rotation prevention portions **350**. However, the present invention is not limited thereto, but the number of the rear rotation prevention portion **350** may be one.

(41) As shown in FIG. 3, each of the rear rotation prevention portions **350** protrudes forward in the front-rear direction from the rear end portion **340**. As shown in FIG. 11, each of the rear rotation prevention portions **350** is positioned at a middle of the housing **300** in the horizontal direction. The two rear rotation prevention portions **350** are arranged in the up-down direction. In the front-rear direction, a front end of each of the rear rotation prevention portions **350** is positioned at a position same as a position of the front end of the wall portion **360**. The two rear rotation prevention portions **350** are positioned at positions same as each other in the right-left direction. The right wall portion **362** is positioned rightward of any of the rear rotation prevention portions **350** in the right-left direction. The right wall portion **362** is coupled with any of the rear rotation prevention portions **350**. The right wall portion **362** couples the two rear rotation prevention portions **350** with each other. The left wall portion **364** is positioned leftward of any of the rear rotation prevention portions **350** in the right-left direction.

(42) The left wall portion **364** is coupled with any of the rear rotation prevention portions **350**. The left wall portion **364** couples the two rear rotation prevention portions **350** with each other. Each of the rear rotation prevention portions **350** is positioned between the first right accommodating portion **372** and the first left accommodating portion **374** in the right-left direction. The first right accommodating portion **372** is positioned rightward of any of the rear rotation prevention portions **350** in the right-left direction. The first left accommodating portion **374** is positioned leftward of any of the rear rotation prevention portions **350** in the right-left direction.

(43) Referring to FIG. 14, the seal member **500** of the present embodiment is formed of elastic material. More specifically, the seal member **500** is formed of rubber. Referring to FIGS. 14 and 15, the seal member **500** has a symmetrical shape with respect to a plane which is perpendicular to the front-rear direction while passing through a middle of the seal member **500** in the front-rear direction. Referring to FIG. 13, the seal member **500** has a symmetrical shape with respect to a plane which is perpendicular to the up-down direction while passing through a middle of the seal member **500** in the up-down direction. The seal member **500** has a symmetrical shape with respect to a plane which is perpendicular to the right-left direction while passing through a middle of the seal member **500** in the right-left direction. When the seal member **500** is viewed along the front-rear direction, the seal member **500** has a closed-ring shape including second sides **520** and second coupling portions **540**. In detail, when the seal member **500** is viewed along the front-rear direction, the seal member **500** has the closed-ring shape including four of the second sides **520** and four of the second coupling portions **540**. Specifically, when the seal member **500** is viewed along the front-rear direction, the seal member **500** has the closed-ring shape which consists of the four second sides **520** and the four second coupling portions **540**. In other words, the seal member **500** has an outer circumference which has a square shape with rounded corners in the plane perpendicular to the front-rear direction.

(44) As shown in FIG. 6, the seal member **500** is attached to the attachment portion **310**. The seal member **500** is surrounded by the surrounding portion **305** in the plane perpendicular to the front-

rear direction. As shown in FIG. 3, in the front-rear direction, a front end of the seal member **500** is positioned rearward of the front end of the surrounding portion **305**.

(45) Referring to FIG. 6, the second sides **520** correspond to the first sides **320**, respectively. Referring to FIGS. 11 and 13, $L_{sub.1} > L_{sub.2}$, where $L_{sub.2}$ is a length of each of the second sides **520** in a state before the seal member **500** is attached to the attachment portion **310**, and $L_{sub.1}$ is a length of the first side **320** corresponding to the second side **520**. Thus, the waterproof connector **100** of the present embodiment can prevent twisting of the seal member **500** when the waterproof connector **100** is mated with or removed from the mating connector. The lengths $L_{sub.2}$ of the second sides **520** are equal to each other in the state before the seal member **500** is attached to the attachment portion **310**. The length $L_{sub.2}$ of each of the second sides **520** in the state before the seal member **500** is attached to the attachment portion **310** is smaller than a length $L_{sub.1B}$ of any of the short sides **324** in the up-down direction. In other words, the waterproof connector **100** fulfills the following inequality: $L_{sub.2} < L_{sub.1B} < L_{sub.1A}$.

(46) As shown in FIG. 6, the second sides **520** include two specific second sides **530** and two second sides **560** other than the specific second sides **530**. Hereinafter, the two second sides **560** other than the specific second sides **530** are also referred to as unspecific second sides **560**. Each of the second sides **560** other than the specific second side **530** has a first flat surface **562**. The first flat surface **562** extends in the plane perpendicular to the front-rear direction and faces rearward in the front-rear direction.

(47) Referring to FIGS. 6 and 13, each of the specific second sides **530** of the present embodiment extends in a first predetermined direction perpendicular to the front-rear direction. In the present embodiment, the first predetermined direction is the horizontal direction. The specific second sides **530** correspond to the two long sides **322**, respectively, of the first sides **320**. Referring to FIGS. 6, 11 and 13, an expected amount G of distortion of each of the second sides **520** upon the attachment of the seal member **500** to the attachment portion **310** is defined by an equation of $G = (L_{sub.1} - L_{sub.2}) / L_{sub.2}$, and the expected amount G of the distortion of each of the specific second sides **530** is maximum among the expected amounts G of the distortions of the second sides **520**.

(48) As understood from FIGS. 6 and 11, a part of each of the specific second sides **530** is accommodated in the first accommodating portion **370**.

(49) Referring to FIG. 14, each of the specific second sides **530** is provided with a front recess **532** and a rear recess **534**.

(50) As shown in FIG. 14, the front recess **532** of the present embodiment is recessed rearward in the front-rear direction. The front recess **532** is positioned at the middle of the seal member **500** in the horizontal direction. The front recess **532** is positioned forward in the front-rear direction beyond the rear recess **534**.

(51) As shown in FIG. 14, the rear recess **534** of the present embodiment is recessed forward in the front-rear direction. The rear recess **534** is positioned at the middle of the seal member **500** in the horizontal direction. The rear recess **534** is positioned rearward in the front-rear direction beyond the front recess **532**. As shown in FIG. 3, the rear rotation prevention portion **350** is positioned within the rear recess **534**.

(52) As shown in FIG. 6, each of the unspecific second sides **560** of the present embodiment extends in a second predetermined direction perpendicular to the front-rear direction. In other words, each of the second sides **560** other than the specific second sides **530** extends in the second predetermined direction perpendicular to the front-rear direction. In the present embodiment, the second predetermined direction is the up-down direction. The unspecific second sides **560** correspond to the two short sides **324**, respectively, of the first sides **320**.

(53) As understood from FIGS. 6 and 11, a part of each of the unspecific second sides **560** is accommodated in the first accommodating portion **370**.

(54) As described above, the first accommodating portion **370** accommodates both of the part of

each of the specific second sides **530** and the part of each of the unspecific second sides **560**. In other words, the first accommodating portion **370** accommodates a part of the seal member **500**. This prevents twisting of the seal member **500**. Additionally, this also prevents the seal member **500** from being moved outward in the direction perpendicular to the front-rear direction when the waterproof connector **100** is mated with the mating connector.

(55) As shown in FIG. **13**, the second coupling portions **540** extend in directions which are different from one another. Each of the second coupling portions **540** couples the second sides **520**, which are adjacent to each other, with each other. Each of the second coupling portions **540** couples the specific second side **530** and the unspecific second side **560**, which are adjacent to each other, with each other. As understood from FIGS. **6** and **11**, a part of each of the second coupling portions **540** is accommodated in the first accommodating portion **370**. As shown in FIG. **6**, the second coupling portions **540** correspond to the first coupling portions **330**, respectively. Each of the second coupling portions **540** is positioned at a position same as a position of the corresponding first coupling portion **330** when the seal member **500** is attached to the attachment portion **310**. Each of the second coupling portions **540** has a second flat surface **542**. The second flat surface **542** extends in the plane perpendicular to the front-rear direction and faces rearward in the front-rear direction. The first flat surface **562** couples the second flat surfaces **542** of the second coupling portions **540**, which are adjacent to each other, with each other.

(56) As described above, the waterproof connector **100** of the present embodiment is configured as follows: the second sides **520** include the two specific second sides **530**; the expected amount G of the distortion of each of the second sides **520** upon the attachment of the seal member **500** to the attachment portion **310** is defined by the equation of $G=(L_{\text{sub.1}}-L_{\text{sub.2}})/L_{\text{sub.2}}$; and the expected amount G of the distortion of each of the specific second sides **530** is maximum among the expected amounts G of the distortions of the second sides **520**. In other words, when the seal member **500** is attached to the attachment portion **310**, the distortion produced in the specific second side **530** is greater than the distortion produced in the unspecific second side **560**. Thus, if the waterproof connector **100** comprises a seal member **500** whose specific second side **530** is provided with none of the front recess **532** and the rear recess **534**, tensional force generated by the specific second side **530** is greater than tensional force, which is generated by the unspecific second side **560**, at a time just after the seal member **500** is attached to the attachment portion **310**. In this case, a position of the second coupling portion **540** on the attachment portion **310** is deviated from the corresponding first coupling portion **330** so that an imbalance between these tensional forces is relieved. This deviation of the position of the second coupling portion **540** degrades sealing characteristics of the waterproof connector **100**.

(57) In contrast, the seal member **500** of the present embodiment is configured so that each of the specific second sides **530** is provided with the front recess **532** and the rear recess **534**. Thus, the waterproof connector **100** of the present embodiment is configured as follows: each of the specific second sides **530**, whose expected amount G of the distortion is maximum among the expected amounts G of the distortions of the second sides **520**, is easily stretched as compared to any of the unspecific second sides **560** whose expected amount G of the distortion is smaller than the expected amount G of the distortion of the specific second side **530**; and an imbalance between tensional force, which is generated by the specific second side **530**, and tensional force, which is generated by the unspecific second side **560**, is relieved at a time just after the seal member **500** is attached to the attachment portion **310**. This prevents a position of the second coupling portion **540** on the attachment portion **310** from being deviated from the corresponding first coupling portion **330**, namely, an appropriate position, when the seal member **500** is attached to the attachment portion **310**. In other words, the waterproof connector **100** of the present embodiment prevents degradation of sealing characteristics.

(58) Referring to FIGS. **13**, **14** and **15**, the waterproof connector **100** is configured as follows: each of the specific second sides **530** has a part which is provided with the front recess **532**; the part of

each of the specific second sides **530** has a cross-section perpendicular to the first predetermined direction; each of the unspecific second side **560** has a cross-section perpendicular to the second predetermined direction; and the cross-section of the part of each of the specific second sides **530** is smaller than the cross-section of any of the unspecific second sides **560**. In other words, the cross-section of the part of each of the specific second sides **530** is smaller than the cross-section of any of the second sides **560** other than the specific second sides **530**. Thus, each of the specific second sides **530** is easily stretched as compared to any of the unspecific second sides **560**, and thereby the position of the second coupling portion **540** on the attachment portion **310** is prevented from being deviated from the appropriate position. In other words, the waterproof connector **100** of the present embodiment prevents degradation of the sealing characteristics.

(59) Referring to FIGS. **13**, **14** and **15**, the waterproof connector **100** is configured as follows: each of the specific second sides **530** has a portion which is provided with the rear recess **534**; the portion of each of the specific second sides **530** has a cross-section perpendicular to the first predetermined direction; and the cross-section of the portion of each of the specific second sides **530** is smaller than the cross-section of any of the unspecific second sides **560**. In other words, the cross-section of the portion of each of the specific second sides **530** is smaller than the cross-section of any of the second sides **560** other than the specific second sides **530**. Thus, each of the specific second sides **530** is easily stretched as compared to any of the unspecific second sides **560**, and thereby the position of the second coupling portion **540** on the attachment portion **310** is prevented from being deviated from the appropriate position. In other words, the waterproof connector **100** of the present embodiment prevents degradation of the sealing characteristics.

(60) As shown in FIGS. **14** and **15**, the seal member **500** has a front accommodated portion **570** and a rear accommodated portion **580**.

(61) As shown in FIG. **14**, the front accommodated portion **570** of the present embodiment is positioned at the front end of the seal member **500** in the front-rear direction. The front accommodated portion **570** includes a front right accommodated portion **572** and a front left accommodated portion **574**.

(62) As shown in FIG. **14**, the front right accommodated portion **572** of the present embodiment is positioned rightward of the front left accommodated portion **574** in the right-left direction. The front right accommodated portion **572** and the front left accommodated portion **574** are arranged in the right-left direction.

(63) As shown in FIG. **14**, the front left accommodated portion **574** of the present embodiment is positioned leftward of the front right accommodated portion **572** in the right-left direction. The front recess **532** is positioned between the front right accommodated portion **572** and the front left accommodated portion **574** in the right-left direction. The front right accommodated portion **572**, the front left accommodated portion **574** and the front recess **532** are positioned at positions same as each other in the front-rear direction.

(64) As shown in FIG. **14**, the rear accommodated portion **580** of the present embodiment is positioned at the rear end **510** of the seal member **500** in the front-rear direction. As understood from FIGS. **6** and **11**, the rear accommodated portion **580** is accommodated in the first accommodating portion **370**. The rear accommodated portion **580** includes a rear right accommodated portion **582** and a rear left accommodated portion **584**.

(65) As shown in FIG. **14**, the rear right accommodated portion **582** of the present embodiment is positioned rightward of the rear left accommodated portion **584** in the right-left direction. The rear right accommodated portion **582** and the rear left accommodated portion **584** are arranged in the right-left direction. As shown in FIG. **3**, the rear right accommodated portion **582** is accommodated in the first right accommodating portion **372**.

(66) As shown in FIG. **14**, the rear left accommodated portion **584** of the present embodiment is positioned leftward of the rear right accommodated portion **582** in the right-left direction. The rear recess **534** is positioned between the rear right accommodated portion **582** and the rear left

accommodated portion **584** in the right-left direction. The rear right accommodated portion **582**, the rear left accommodated portion **584** and the rear recess **534** are positioned at positions same as each other in the front-rear direction. As shown in FIG. 3, the rear left accommodated portion **584** is accommodated in the first left accommodating portion **374**.

(67) Referring to FIG. 7, each of the terminals **200** of the present embodiment is made of metal. Each of the terminals **200** is a so-called female contact. As shown in FIG. 6, the terminals **200** are held by the housing **300**.

(68) Referring to FIG. 7, the waterproof connector **100** further comprises a front retainer **400**.

(69) As shown in FIG. 4, the front retainer **400** of the present embodiment is attached to the attachment portion **310**. As shown in FIG. 16, the front retainer **400** has a front end portion **480**.

(70) As shown in FIG. 3, the front end portion **480** of the present embodiment faces rearward in the front-rear direction. The seal member **500** is sandwiched between the front end portion **480** and the rear end portion **340** in the front-rear direction. In other words, the seal member **500** is sandwiched between the front retainer **400** and the housing **300** in the front-rear direction.

(71) As shown in FIG. 16, the front retainer **400** further has a second accommodating portion **470**.

(72) Referring to FIG. 16, the second accommodating portion **470** is a recess which is recessed forward in the front-rear direction. The front end portion **480** defines a front end of the second accommodating portion **470** in the front-rear direction. Referring to FIGS. 3, 13 and 16, the second accommodating portion **470** accommodates a part of each of the specific second sides **530** and a part of each of the unspecific second sides **560**. In other words, the second accommodating portion **470** accommodates a part of the seal member **500**. This further prevents twisting of the seal member **500**. Additionally, this also further prevents the seal member **500** from being moved outward in the direction perpendicular to the front-rear direction when the waterproof connector **100** is mated with the mating connector. The second accommodating portion **470** includes a second right accommodating portion **472** and a second left accommodating portion **474**.

(73) As shown in FIG. 16, the second right accommodating portion **472** of the present embodiment is positioned rightward of the second left accommodating portion **474** in the right-left direction. The second right accommodating portion **472** and the second left accommodating portion **474** are arranged in the right-left direction. As shown in FIG. 3, the front right accommodated portion **572** is accommodated in the second right accommodating portion **472**.

(74) As shown in FIG. 16, the second left accommodating portion **474** of the present embodiment is positioned leftward of the second right accommodating portion **472** in the right-left direction. As shown in FIG. 3, the front left accommodated portion **574** is accommodated in the second left accommodating portion **474**.

(75) As shown in FIG. 16, the front retainer **400** of the present embodiment is provided with two front rotation prevention portions **410**. However, the present invention is not limited thereto, but the number of the front rotation prevention portion **410** may be one.

(76) Referring to FIG. 3, each of the front rotation prevention portions **410** of the present embodiment protrudes in the front-rear direction from the front end portion **480**. More specifically, each of the front rotation prevention portions **410** protrudes rearward in the front-rear direction from the front end portion **480**. The front rotation prevention portion **410** is positioned within the front recess **532**. As shown in FIG. 4, in the front-rear direction, a rear end of each of the front rotation prevention portions **410** is positioned rearward beyond the front end of the surrounding portion **305** of the housing **300**.

(77) As shown in FIG. 16, each of the front rotation prevention portions **410** is positioned between the second right accommodating portion **472** and the second left accommodating portion **474** in the right-left direction. The second right accommodating portion **472** is positioned rightward of any of the front rotation prevention portions **410** in the right-left direction. The second left accommodating portion **474** is positioned leftward of any of the front rotation prevention portions **410** in the right-left direction.

(78) Although the specific explanation about the present invention is made above referring to the embodiments, the present invention is not limited thereto and is susceptible to various modifications and alternative forms.

(79) Although the waterproof connector **100** of the aforementioned embodiment is configured so that the second sides **520** include the two specific second sides **530**, the present invention is not limited thereto. Specifically, the second sides **520** should include at least one specific second side **530**. In this case, the waterproof connector **100** should be configured as follows: the expected amount G of the distortion of each of the second sides **520** upon the attachment of the seal member **500** to the attachment portion **310** is defined by the equation of $G=(L_{sub.1}-L_{sub.2})/L_{sub.2}$; the expected amount G of the distortion of the at least one specific second side **530** is maximum among the expected amounts G of the distortions of the second sides **520**; and the at least one specific second side **530** is provided with the front recess **532** and the rear recess **534**. This enables the waterproof connector **100** to obtain the same effect as in the present embodiment.

(80) Although the housing **300** of the aforementioned embodiment has the rear end portion **340**, the first accommodating portion **370** and the rear rotation prevention portions **350**, the present invention is not limited thereto. Specifically, the housing **300** may have no first accommodating portion **370**. Additionally, the housing **300** may have no rear rotation prevention portion **350**. Similarly, although the front retainer **400** of the aforementioned embodiment has the front end portion **480**, the second accommodating portion **470** and the front rotation prevention portions **410**, the present invention is not limited thereto. Specifically, the front retainer **400** may have no second accommodating portion **470**. Additionally, the front retainer **400** may have no front rotation prevention portion **410**. In order to obtain the same effect, which is that the seal member **500** is sandwiched between the front retainer **400** and the housing **300** in the front-rear direction, as in the present embodiment, the waterproof connector **100** is required to be configured so that the housing **300** has the rear end portion **340** while the front retainer **400** has the front end portion **480**. In the waterproof connector **100** with such a configuration, the rear rotation prevention portion **350** is required to protrude forward in the front-rear direction from the rear end portion **340** if the housing **300** is provided with the rear rotation prevention portion **350**. Similarly, in the waterproof connector **100** with such a configuration, the front rotation prevention portion **410** is required to protrude rearward in the front-rear direction from the front end portion **480** if the front retainer **400** is provided with the front rotation prevention portion **410**.

(81) While there has been described what is believed to be the preferred embodiment of the invention, those skilled in the art will recognize that other and further modifications may be made thereto without departing from the spirit of the invention, and it is intended to claim all such embodiments that fall within the true scope of the invention.

Claims

1. A waterproof connector mateable with a mating connector along a front-rear direction, wherein: the waterproof connector comprises a terminal, a housing and a seal member; the terminal is held by the housing; the housing is provided with an attachment portion; when the attachment portion is viewed along the front-rear direction, the attachment portion has an outer circumferential shape including first sides and first coupling portions; the first coupling portions extend in directions which are different from one another; each of the first coupling portions couples the first sides, which are adjacent to each other, with each other; the seal member is formed of elastic material and is attached to the attachment portion; when the seal member is viewed along the front-rear direction, the seal member has a closed-ring shape including second sides and second coupling portions; the second sides correspond to the first sides, respectively; the second coupling portions extend in directions which are different from one another; each of the second coupling portions couples the second sides, which are adjacent to each other, with each other; the second sides

include at least one specific second side; the at least one specific second side is provided with a front recess and a rear recess; the front recess is recessed rearward in the front-rear direction; the rear recess is recessed forward in the front-rear direction; $L_{sub.1} > L_{sub.2}$, where $L_{sub.2}$ is a length of each of the second sides in a state before the seal member is attached to the attachment portion, and $L_{sub.1}$ is a length of the corresponding first side; an expected amount G of distortion of each of the second sides upon attachment of the seal member to the attachment portion is defined by an equation of $G = (L_{sub.1} - L_{sub.2}) / L_{sub.2}$; the expected amount G of the distortion of the at least one specific second side is maximum among the expected amounts G of the distortions of the second sides; each of the second sides other than the specific second side has a first flat surface; the first flat surface extends in a plane perpendicular to the front-rear direction and faces rearward in the front-rear direction; each of the second coupling portions has a second flat surface; the second flat surface extends in the plane perpendicular to the front-rear direction and faces rearward in the front-rear direction; the first flat surface couples the second flat surfaces of the second coupling portions, which are adjacent to each other, with each other; the at least one specific second side includes two of the specific second sides; the first sides include two long sides and two short sides; each of the two long sides extends in a horizontal direction perpendicular to the front-rear direction; each of the two short sides extends in an up-down direction perpendicular to both the front-rear direction and the horizontal direction; the specific second sides correspond to the two long sides, respectively, of the first sides; a size of the long side in the horizontal direction is greater than a size of the short side in the up-down direction; and the lengths of the second sides are equal to each other in the state before the seal member is attached to the attachment portion.

2. The waterproof connector as recited in claim 1, wherein: the specific second side extends in a first predetermined direction perpendicular to the front-rear direction; each of the second sides other than the specific second side extends in a second predetermined direction perpendicular to the front-rear direction; the specific second side has a part which is provided with the front recess; the part of the specific second side has a cross-section perpendicular to the first predetermined direction; each of the second sides other than the specific second side has a cross-section perpendicular to the second predetermined direction; and the cross-section of the part of the specific second side is smaller than the cross-section of any of the second sides other than the specific second side.

3. The waterproof connector as recited in claim 1, wherein: the housing has a rear end portion; the rear end portion is positioned outside the attachment portion in the plane perpendicular to the front-rear direction; the attachment portion extends forward in the front-rear direction from the rear end portion; the seal member has a rear end in the front-rear direction; and the rear end of the seal member is pressed against the rear end portion.

4. The waterproof connector as recited in claim 3, wherein: the housing is provided with a rear rotation prevention portion; the rear rotation prevention portion protrudes forward in the front-rear direction from the rear end portion; and the rear rotation prevention portion is positioned within the rear recess.

5. The waterproof connector as recited in claim 3, wherein: the waterproof connector further comprises a front retainer; the front retainer is attached to the attachment portion; the front retainer has a front end portion; and the seal member is sandwiched between the front end portion and the rear end portion in the front-rear direction.

6. The waterproof connector as recited in claim 5, wherein: the front retainer is provided with a front rotation prevention portion; the front rotation prevention portion protrudes rearward in the front-rear direction from the front end portion; and the front rotation prevention portion is positioned within the front recess.
